

AWS EC2 Linux Instance & SSH Lab

Hands-on AWS Cloud Lab • Amazon Linux 2023 • EC2 • SSH

Objective

To create a Linux-based Amazon EC2 instance using the AWS Management Console, configure secure SSH access using a key pair and security group, connect from Windows PowerShell, and verify the Linux environment using basic commands.

Environment Used

Cloud	Amazon Web Services (AWS)
Service	Amazon EC2
Region	Asia Pacific (Mumbai) / ap-south-1
Operating System	Amazon Linux 2023
Instance Type	t3.micro
Storage	8 GiB EBS volume
Connection	SSH from Windows PowerShell
SSH User	ec2-user

1. Launch the EC2 Instance

From the AWS EC2 dashboard, select **Launch instance**. Set the instance name to **AWS-EC2-Lab**, select the Amazon Linux 2023 AMI, and use the t3.micro instance type. Create/select the key pair **ec2-lab-key** and securely keep the downloaded private key.

The following screenshots document the AWS console configuration and successful instance launch. Public IP/DNS information has been redacted where appropriate.

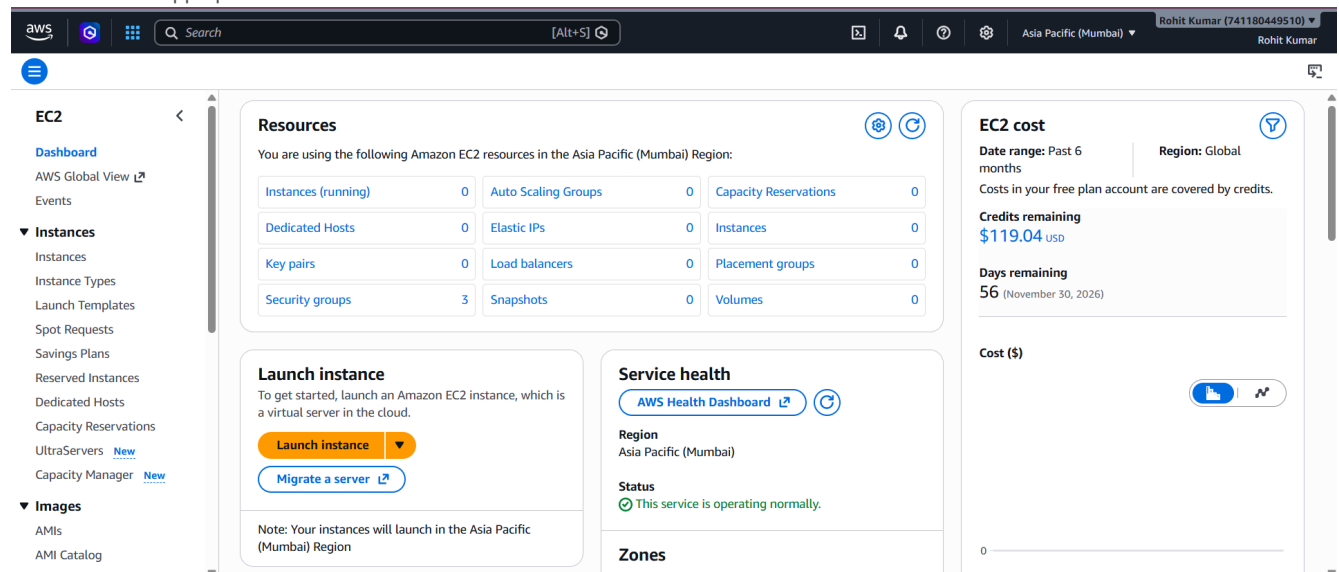


Figure 1. EC2 dashboard before launching the instance.

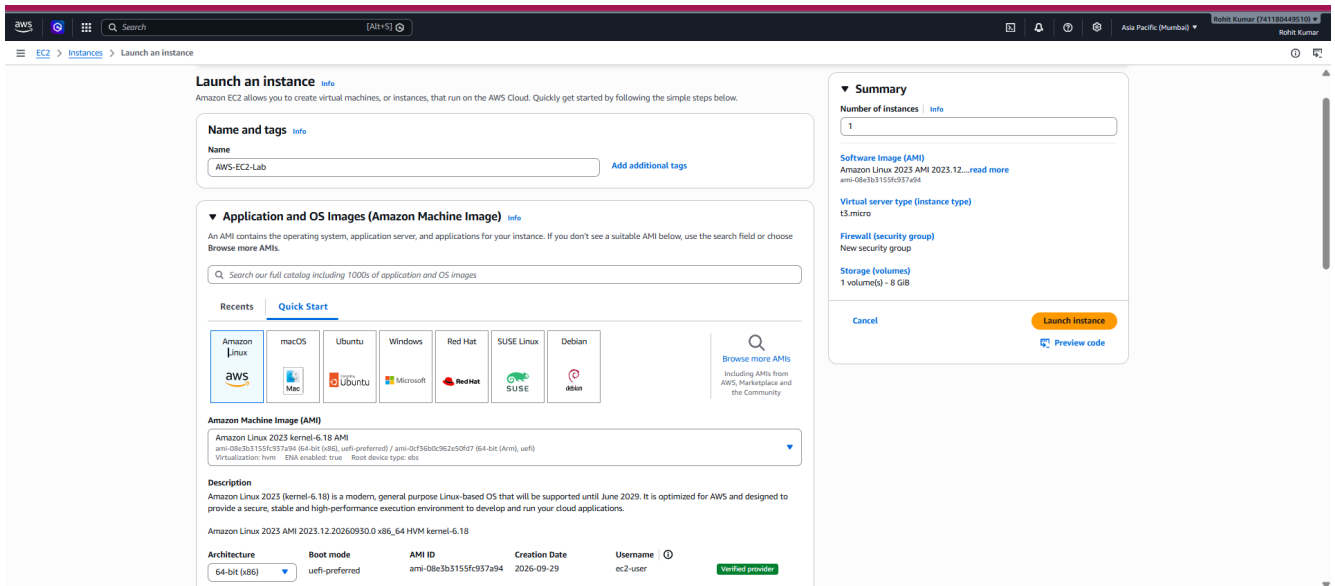


Figure 2. Launch configuration with instance name, Amazon Linux 2023 AMI and architecture.

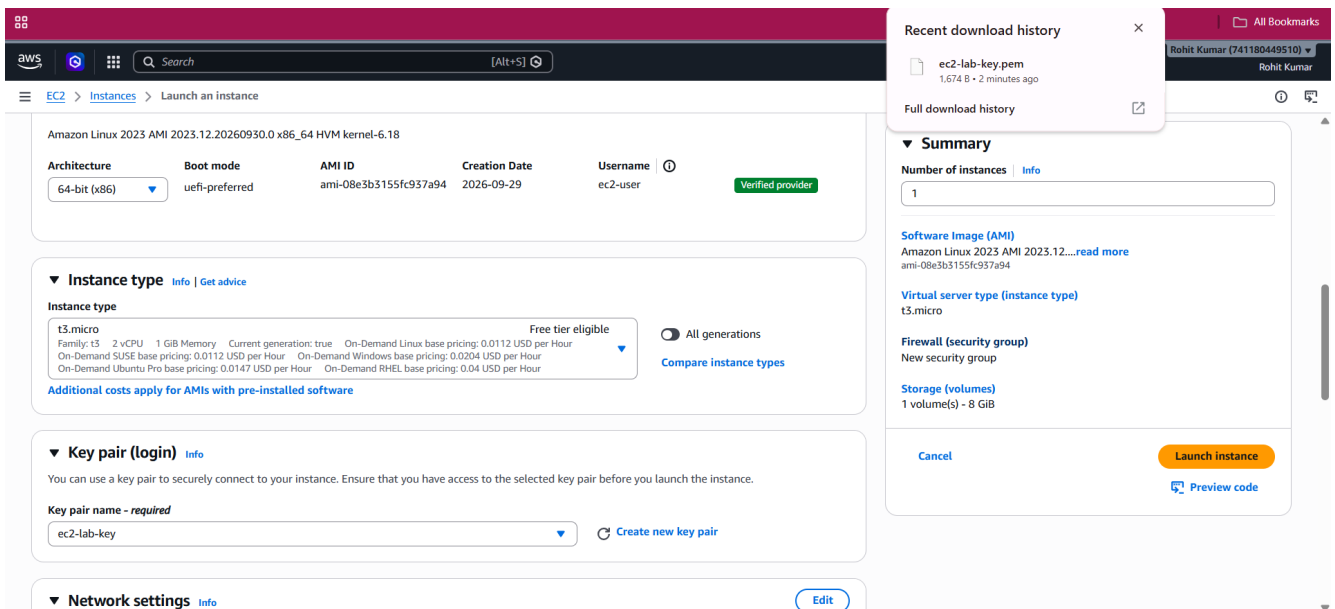


Figure 3. Instance type and SSH key-pair configuration.

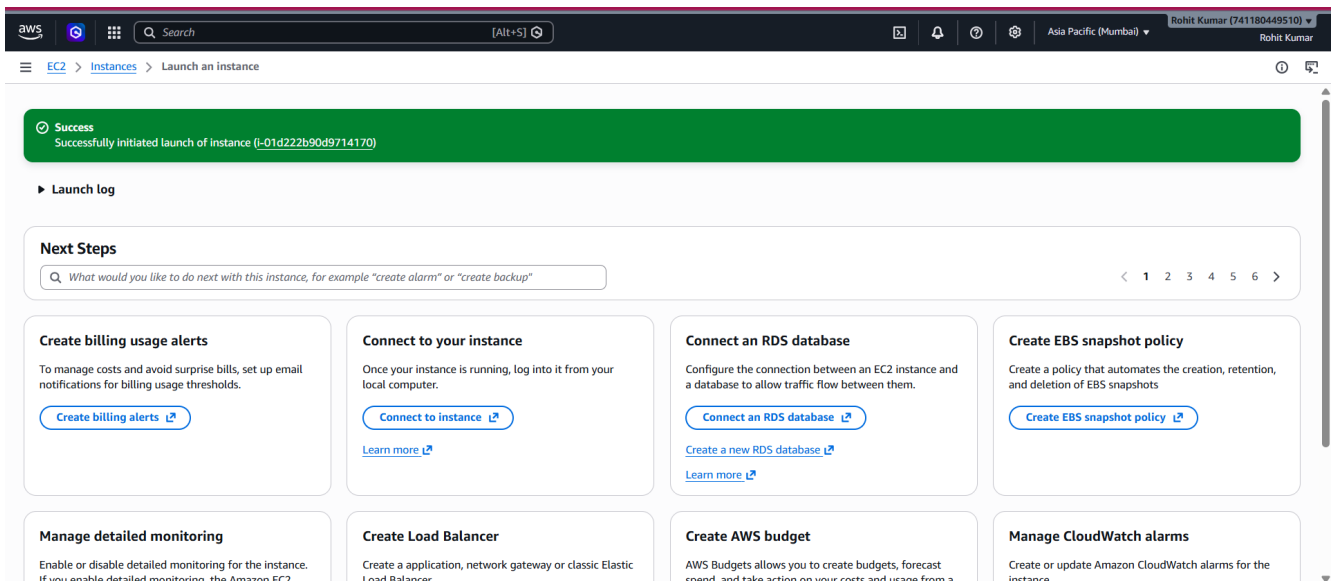


Figure 4. Successful instance launch confirmation.

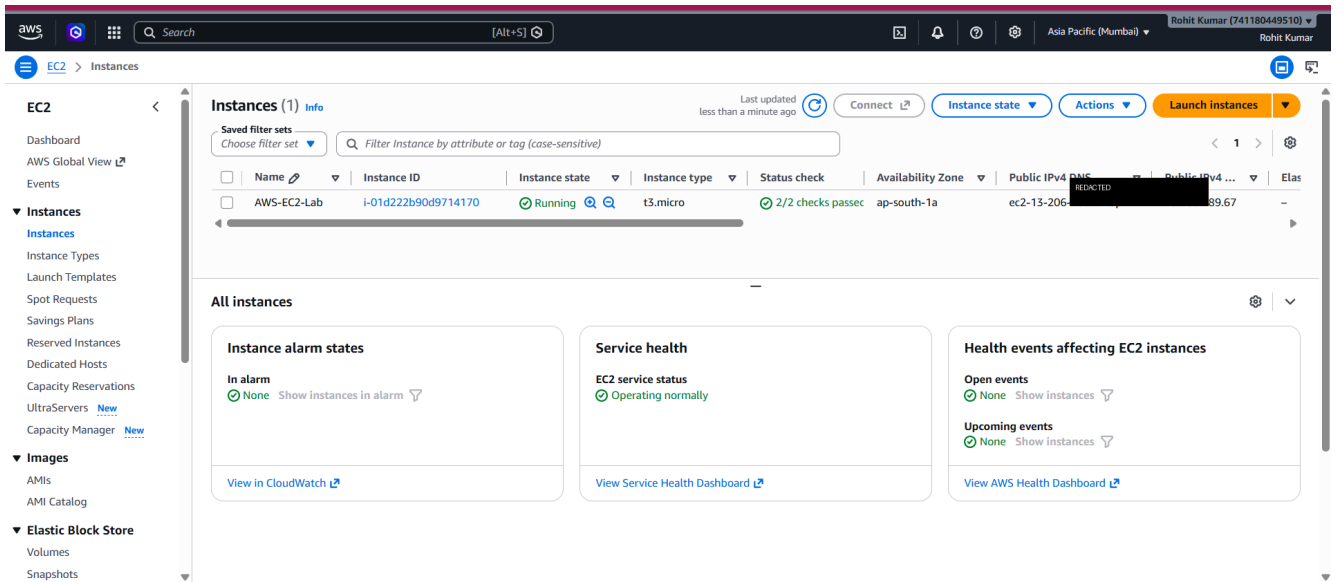


Figure 5. EC2 instance running with 2/2 status checks passed.

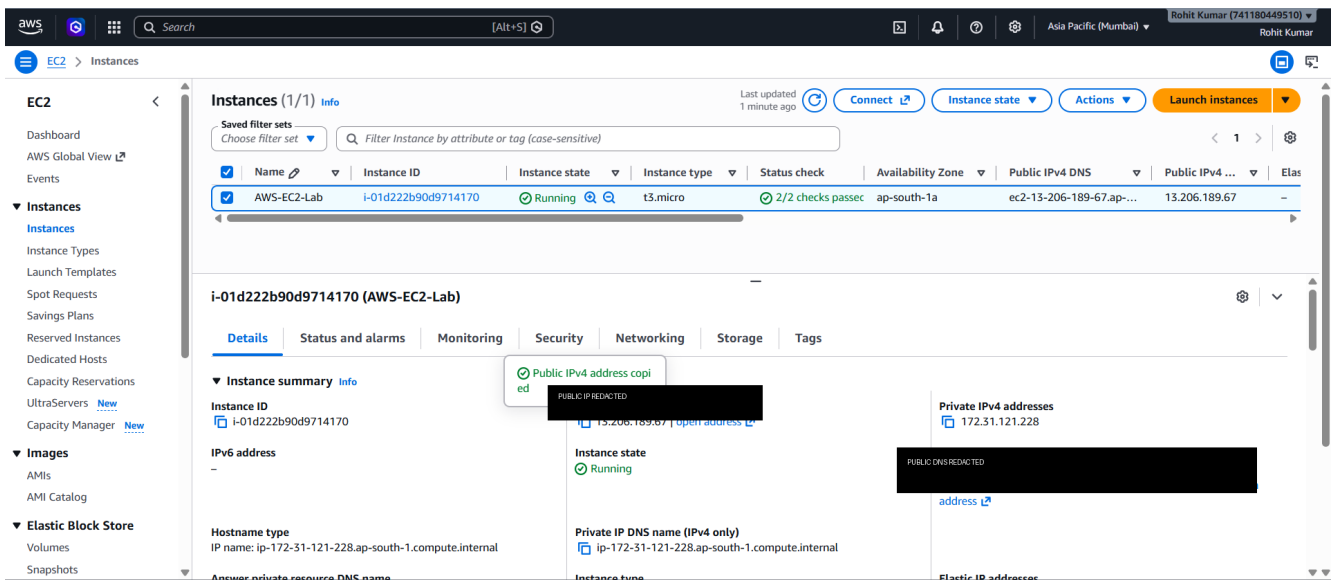


Figure 6. Instance details and networking information.

2. Connect to EC2 Using SSH

After the instance reaches **Running** state and the status checks pass, connect from Windows PowerShell using the downloaded key pair and the instance public IPv4 address.

```
cd Downloads
ssh -i "ec2-lab-key.pem" ec2-user@<PUBLIC-IP>
```

On the first connection, confirm the host authenticity by entering **yes**. A successful connection opens an Amazon Linux shell with the `ec2-user` account.

Security note: Keep the .pem private key confidential. Do not share or expose private keys, passwords, access keys, tokens, or other credentials.

```
ec2-user@ip-172-31-121-228:~$ ssh -i "ec2-user@13.206.16" ec2-user@13.206.16
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\rohit> cd Downloads
PS C:\Users\rohit\Downloads>
PS C:\Users\rohit\Downloads> ssh -i "ec2-lab-key.pem" ec2-user@13.206.16
Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

Last login: Mon Oct 5 11:53:30 2026 from 122.171.23.175
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ whoami
ec2-user
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ hostname
ip-172-31-121-228.ap-south-1.compute.internal
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ cat /etc/os-release
NAME="Amazon Linux"
ID_LIKE="fedora"
VERSION_ID="2023"
PLATFORM_ID="platform:al2023"
PRETTY_NAME="Amazon Linux 2023.12.20260930"
ANSI_COLOR="0;33"
CPE_NAME="cpe:2.3:o:amazon:amazon_linux:2023"
HOME_URL="https://aws.amazon.com/linux/amazon-linux-2023/"
DOCUMENTATION_URL="https://docs.aws.amazon.com/linux/"
SUPPORT_URL="https://aws.amazon.com/premiumsupport/"
BUG_REPORT_URL="https://github.com/amazonlinux/amazon-linux-2023"
VENDOR_NAME="AWS"
VENDOR_URL="https://aws.amazon.com/"
SUPPORT_END="2029-06-30"
```

Figure 7. Successful SSH connection to the Amazon Linux EC2 instance.

3. Verify the Linux Environment

The following commands were executed successfully on the EC2 instance:

```
whoami
hostname
cat /etc/os-release
ip addr
df -h
free -h
```

Observed results: the login user was **ec2-user**; the operating system was **Amazon Linux 2023**; the network interface had a private address in the VPC; disk usage was displayed; and memory information was displayed successfully.

```
ec2-user@ip-172-31-121-228:~$ cat /etc/hostname
HOME_URL="https://aws.amazon.com/linux/amazon-linux-2023/"
DOCUMENTATION_URL="https://docs.aws.amazon.com/linux/"
SUPPORT_URL="https://aws.amazon.com/premiumsupport/"
BUG_REPORT_URL="https://github.com/amazonlinux/amazon-linux-2023"
VENDOR_NAME="AWS"
VENDOR_URL="https://aws.amazon.com/"
SUPPORT_END="2029-06-30"
[ec2-user@ip-172-31-121-228 ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 02:ff:fe:09:2b:65 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-07d7f364181545ac0
    altname device-number-0.0
    inet 172.31.121.228/16 metric 512 brd 172.31.255.255 scope global dynamic ens5
        valid_lft 2897sec preferred_lft 2897sec
    inet6 fe80::ff:fe:09:2b:65/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs         4.0M   0  4.0M   0% /dev
tmpfs            457M   0  457M   0% /dev/shm
tmpfs            183M 444K  183M   1% /run
efivarfs         128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1    8.0G  1.6G   6.4G  20% /
tmpfs            457M   0  457M   0% /tmp
/dev/nvme0n1p128  10M   1.3M   8.7M  13% /boot/efi
tmpfs            92M   0   92M   0% /run/user/1000
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ free -h
              total        used        free      shared  buff/cache   available
Mem:           913Mi       152Mi       514Mi         0.0Ki       245Mi       628Mi
Swap:            0B           0B           0B
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ |
```

Figure 8. Linux network, disk and memory verification.

4. Create and Verify a Test File

A test file was created to demonstrate successful command execution and basic file operations:

```
echo "AWS EC2 Lab Successfully Completed" > lab.txt
cat lab.txt
```

The expected text was displayed, confirming that the file was successfully created and read on the EC2 Linux instance.

```
Windows PowerShell
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 02:ff:fe:09:2b:65 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-07d7f364181545ac0
    altname device-number-0.0
    inet 172.31.121.228/16 metric 512 brd 172.31.255.255 scope global dynamic ens5
        valid_lft 2897sec preferred_lft 2897sec
    inet6 fe80::ff:fe:09:2b:65/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs         4.0M   0  4.0M   0% /dev
tmpfs            457M   0  457M   0% /dev/shm
tmpfs            183M 444K  183M   1% /run
efivarfs         128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1    8.0G  1.6G   6.4G  20% /
tmpfs            457M   0  457M   0% /tmp
/dev/nvme0n1p128  10M   1.3M   8.7M  13% /boot/efi
tmpfs            92M   0   92M   0% /run/user/1000
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ free -h
              total        used        free      shared  buff/cache   available
Mem:           913Mi       152Mi       514Mi         0.0Ki       245Mi       628Mi
Swap:            0B           0B           0B
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ echo "AWS EC2 Lab Successfully Completed" > lab.txt
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ cat lab.txt
AWS EC2 Lab Successfully Completed
[ec2-user@ip-172-31-121-228 ~]$
[ec2-user@ip-172-31-121-228 ~]$ exit
logout
Connection to 13.206.189.67 closed.
PS C:\Users\rohit\Downloads>
```

Figure 9. Test file creation, verification and SSH logout.

5. Key AWS Concepts Demonstrated

EC2	Virtual server/compute service in AWS.
AMI	Template used to create the operating system environment.
Instance Type	Defines compute capacity such as vCPU and memory.
Key Pair	Public/private key mechanism used for SSH authentication.
Security Group	Virtual firewall controlling network traffic.
Public IPv4	Address used to reach the instance when public access is configured.
Private IP	Address used inside the VPC/private network.
EBS	Persistent block storage attached to the EC2 instance.

6. Result

The Linux-based Amazon EC2 instance was successfully created using the AWS Management Console. SSH access was configured and a secure connection was established from Windows PowerShell. Basic Linux commands, networking checks, storage checks, memory checks, and file operations were successfully performed on the EC2 instance.

7. Conclusion

This experiment demonstrated the basic AWS EC2 provisioning workflow and remote administration of a Linux virtual machine using SSH. It also provided practical exposure to Amazon Linux, security groups, key-pair authentication, EBS storage, and common Linux commands.